

Trust in AI news, AI literacy, and the mediating role of artificial intelligence attitudes: A longitudinal study across diverse societies

Manuel Goyanes^a, Sonja Utz^{b,f}, Homero Gil de Zúñiga^{c,d,e,*}

^a MITCOM Research Group, Department of Communication, Carlos III University of Madrid, Madrid, Spain

^b Department of Psychology, University of Tübingen, Tübingen, Germany

^c Democracy Research Unit, Political Science, College of Law & Public Administration, University of Salamanca, Salamanca, Spain

^d Media Effects Research Lab, Film Production & Media Studies Department, Donald P. Bellisario College of Communications, Pennsylvania State University, University Park, PA, USA

^e Facultad de Comunicación y Letras, Universidad Diego Portales, Santiago, Chile

^f Everyday Media Lab, Leibniz-Institut für Wissensmedien, Tübingen, Germany

ARTICLE INFO

Keywords:

Artificial intelligence
Trust in news
Artificial intelligence news trust
Artificial intelligence literacy
Artificial intelligence attitudes

ABSTRACT

The rise of generative artificial intelligence (AI) tools, such as ChatGPT, has introduced new ways for people to access and interact with news. Yet, little is known about the factors that shape citizens' trust in AI-generated news. Drawing on a two-wave cross-national panel survey conducted in the United States ($n = 1815$), Spain ($n = 1811$), and Chile ($n = 1802$), this study investigates how AI literacy and attitudes toward AI influence trust in AI-generated news over time. We propose and test a mediation model in which AI literacy affects news trust indirectly through attitudes toward AI. Results show that while AI literacy does not exert a direct influence on trust in AI-generated news, it fosters more positive attitudes toward AI, which in turn enhance trust. This indirect effect is consistent across all three countries, suggesting an attitudinal mechanism linking literacy and trust in AI news trust. Overall, our findings highlight the key role of positive attitudinal evaluations in shaping public trust in communicative AI and underscore that fostering trust requires both improving citizens' understanding of AI and addressing their broader attitudes toward this technology.

Generative AI tools, such as ChatGPT, have gained rapid popularity and are increasingly used for news and information searches; information search is meanwhile the top reason for using generative AI and (although small) the percentage of people using AI for news has doubled from 2024 to 2025 (Simon et al., 2025). While relying on these tools may pose more limited challenges for long and well-established facts, concerns arise regarding their use for news and current events. Although newer models now access the internet, their ability to source information accurately remains inconsistent (Jaźwińska & Chandrasekar, 2025). Since large language models (LLMs) generate responses based on statistical probability, there is always a risk of generating hallucinated information (Tauman Kalai et al., 2025). Nevertheless, interacting with generative AI could potentially facilitate easier access to news, as these tools can tailor content to users' individual interests or explain topics in more accessible language.

It is, thus, important to understand whether people trust news delivered by AI and to identify the factors that predict such trust. AI

represents a relatively new technology and differs fundamentally from traditional and social media in its capacity to function not as a mere channel but as an interaction partner (Guzman & Lewis, 2020). Accordingly, services such as ChatGPT, CoPilot, and Siri are frequently described as communicative AI. Within this context, one important factor in using AI for news is AI literacy, which we operationalize as the ability to understand and effectively utilize AI technologies (Carolus et al., 2023; Long & Magerko, 2020; Ng et al., 2021). We argue that higher levels of AI literacy facilitate informed engagement with AI-driven news content. General attitudes toward AI might also play a significant role, as positive attitudes have consistently been linked to greater trust in technology and serve as key predictors of technology acceptance (Marikyan et al., 2023).

The goal of this paper is to advance understanding of trust in AI-generated news. To achieve this, we draw on data from a longitudinal panel study conducted in three countries: Spain, the United States, and Chile. While much scholarship on media trust focuses on WEIRD

* Corresponding author. Facultad de Comunicación y Letras, Universidad Diego Portales, Santiago, Chile.

E-mail address: hg2@usal.es (H. Gil de Zúñiga).

(Western, Educated, Industrialized, Rich, and Democratic) contexts, the inclusion of Chile, representing the Global South, allows for examination of the generalizability of findings across diverse cultural settings and helps determine whether observed processes are culturally contingent. Furthermore, the use of a longitudinal design offers richer insights than a cross-sectional approach, as it enables the tracking of attitudinal and behavioral changes over time and captures the evolving dynamics of trust in AI news. Our results show indirect effects of AI literacy on trust in AI-generated news, mediated via attitudes towards AI. This pattern was the same in all three countries.

1. Theoretical framework

Trust in media and other technologies is a key variable in the realm of media effect as it predicts both whether and how individuals engage with a given medium (Schranz et al., 2018). For instance, higher trust in news media has been shown to reduce susceptibility to misinformation and fake news (Ryan, 2021). Generally, trust in media is closely linked to the broader concept of media credibility, which is predominantly shaped by two core components: expertise and trustworthiness (Kohring, 2019; Strömbäck et al., 2020). The question whether people trust (news) media has received more attention in the last decades when the Internet and social media started to provide a high-choice media environment and it turned out that some people do not trust traditional news media anymore (Strömbäck et al., 2020).

Communicative AI is not just another channel like social media. It has fundamentally transformed the way individuals interact with technology, serving as an interaction partner rather than merely a communication channel (Guzman, 2020; Lombard & Xu, 2021). Unlike traditional web browsing or social media consumption where users passively read content, communicative AI requires active engagement through prompts, fostering conversational exchanges. While search engines like Google have long allowed query formulation as questions, their responses are limited to presenting links rather than engaging in dialogue. This conversational format enhances accessibility, enabling most users to interact effortlessly through chat-based interfaces. Additionally, AI's ability to personalize news and provide simplified explanations makes it a valuable tool for information consumption (Sultan et al., 2024). However, important caveats exist, especially concerning news defined by its timeliness, because these models often lack accuracy in source attribution (Jaźwińska & Chandrasekar, 2025), and the information generated may reflect biases depending on the LLM employed. For instance, the Chinese LLM DeepSeek exhibits reluctance to address politically sensitive topics such as the Tiananmen Square events or Taiwan (Lu, 2025).

As of today, research on AI-generated news remains limited. One survey conducted across ten countries, which primarily examined algorithmic news recommendations, found that people generally exhibited a slight preference for human over AI-generated news recommendation and content creation (Araujo et al., 2023). Notably, younger individuals and men demonstrated greater openness toward AI-generated news, and higher levels of institutional trust were positively associated with acceptance of AI-based news recommendations. Gondwe (2025) investigated perceptions of AI-generated news in ten African countries, revealing generally neutral attitudes toward such content, with younger demographics again showing greater receptiveness. Regarding psychological processes, Gondwe (2025) speculated about the role of digital literacy in fostering trust in AI news, suggesting that individuals with higher literacy levels would be better equipped to critically evaluate AI-generated content. Digital or AI literacy was, however, not measured in the study. We aim to fill this gap and examine the relationship between AI literacy and trust in AI-generated news.

1.1. AI literacy and AI news trust

AI literacy can be considered as a specific form of media or digital

literacy (Valtonen et al., 2019). Current available literature defined AI literacy in different ways. Ng et al. (2021) conducted an exploratory review and identified four core components: knowing and understanding, using and applying, evaluating and creating, and addressing ethical issues. According to Carolus et al. (2023), most conceptualizations emphasize not in-depth technical knowledge but rather the ability to critically evaluate AI and use it efficiently. A higher AI literacy is often considered as a factor that fosters more beneficial outcomes of AI use such as more critical evaluation of AI output or a more efficient and ethical usage (Long & Magerko, 2020; Pinski & Benlian, 2024). Empirical studies show, however, a more complex pattern.

When it comes to the relationship between AI literacy and trust in AI-generated news, two relationships are possible. On one hand, higher AI literacy may increase awareness of potential biases and hallucinations in AI outputs, thus lowering trust in AI-generated news. Supporting this view, Tully et al. (2025) found that individuals with lower AI literacy tended to be more receptive to AI, partly because they perceived it as more magical. Moreover, literate users may fact-check AI-generated content more frequently. In fact, positive associations between media literacy and fact-checking behaviors have been documented for health information consumption on websites and social media (Lee & Ramazan, 2021).

On the other hand, higher literacy may enhance users' ability to formulate effective prompts, leading to better AI-generated responses. In line with this, recent research reported differential relationships between facets of AI literacy and verification behaviors. Rheu and Cho (2025) found that understanding the principles of LLMs correlated positively with verification behaviors but that self-efficacy and knowledge of LLM features correlated negatively with verification behaviors; people instead relied more on the machine heuristic and assumed that LLMs are accurate and objective (Sundar & Kim, 2019).

Similar complex relationships between literacy and trust have been described for news media use in general. Schwarzenegger (2020) found in a qualitative study that people with low knowledge about the media system expressed so-called naïve pragmatic trust in the media. But people who were skeptical about news media because they were aware that news is made by humans and/or influenced by external variables (i. e., who had higher literacy about the media system) still showed informed pragmatic trust in certain news sources. Thus, higher literacy can come with higher trust, even if one is more aware about shortcomings.

Positive correlations between AI literacy and trust have also been reported in contexts such as human-AI team collaboration (Pan et al., 2025) and across multiple practical AI applications (Huang & Ball, 2024). In this paper, we focus more on the skills and efficacy component of AI literacy. Based on the findings by Rheu and Cho (2025), we hypothesize a positive relationship between AI literacy and subsequent trust in AI news:

H1). Artificial intelligence literacy at W^1 is positively associated with artificial intelligence news trust at W^2 in a) the pooled sample, b) the USA, c) Spain and d) Chile

1.2. Attitudes towards AI and AI news trust

Many individuals lack a deep understanding of the technical foundations of AI models and thus score low on AI literacy (Hornberger et al., 2025). Instead of relying on knowledge and comprehension, they may base their trust in AI-generated news on their attitudes. Attitudes, defined as evaluations of an object or concept (Vogel & Wanke, 2016), have been shown to play a crucial role in technology acceptance (Davis, 1989). Negative attitudes toward AI are often considered significant barriers to its adoption and trustworthiness (De Freitas et al., 2023). In the domain of trust in news media, Schwarzenegger (2020) identified a second, more sentiment-based path to trust in media, that is often mentioned by people who do not understand the news system

completely. A meta-analysis documented a positive relationship between attitudes toward technology and trust in that technology (Marikyan et al., 2023), a finding echoed specifically in studies of attitudes toward and trust in ChatGPT (Montag & Ali, 2025). Based on these findings, we expect a similar positive relationship in the context of news-related AI use and predict:

H2). Favorable attitudes toward artificial intelligence at W^1 are positively associated with artificial intelligence news trust at W^2 in a) the pooled sample, b) the USA, c) Spain and d) Chile

1.3. AI literacy and AI attitudes

Several studies have reported positive correlations between AI literacy and attitudes toward AI (Çayak, 2024; Laupichler et al., 2024). However, less attention has been given to whether this relationship is causal and what the direction of the effect might be. It is less plausible to assume that attitudes necessarily lead to higher literacy. Although positive attitudes may encourage AI use, this does not guarantee increased knowledge about AI. Rather, it seems more plausible that individuals with greater competence in using AI develop more positive attitudes because they perceive AI as easier to use and find its outputs more useful, accurate and unbiased (Rheu & Cho, 2025). According to the Technology Acceptance Model (Davis, 1989), both perceived ease of use and perceived usefulness are key determinants of attitudes toward technology. This assumption is supported by a qualitative study showing that attitudes toward AI become more positive with increased experience and knowledge (Daly et al., 2025). It stands to reason that as people develop AI literacy, they will also develop more favorable AI attitudes in time, even when controlling subjects' prior attitudes towards AI with our longitudinal data. More formally:

H3). Artificial intelligence literacy at W^1 is positively associated with favorable attitudes toward artificial intelligence at W^2 in a) the pooled sample, b) the USA, c) Spain and d) Chile

1.4. Artificial intelligence literacy, attitudes toward artificial intelligence, and artificial intelligence trust

Additionally, we expect that attitudes toward AI mediate the relationship between AI literacy and trust in AI-generated news. Prior work has found that people with higher AI literacy are more aware of potential biases or less inclined to view AI as an infallible source (Tully et al., 2025). However, a higher awareness of bias does not always come with lower trust; Gondwe (2025) found a positive correlation between these two variables. We aim to explain why higher AI literacy could come with higher trust in AI. Building on Rheu and Cho (2025), we argue that especially the efficacy dimension of AI literacy could be related to higher trust in AI. We assume that people who report higher skills in using AI are better at prompting AI and able to use it more effectively and efficiently. As a consequence of these more positive experiences, they develop more positive attitudes towards AI which in turn foster greater trust in AI-generated news. Therefore, we hypothesize that the positive effect of AI literacy on trust in AI news is indirectly influenced by individuals' attitudes toward AI (see Fig. 1).

H4). Artificial intelligence literacy at W^1 is positively associated with favorable attitudes toward artificial intelligence at W^1 , which in turn are positively associated with increased trust in artificial intelligence news at W^2 in a) the pooled sample, b) the USA, c) Spain and d) Chile

2. Method

The data analyzed in this study was collected from a multi-wave cross-national survey administered under the oversight of the [Anonymized] University Institutional Review Board (Protocol #1282). The project was part of a funded international initiative examining media

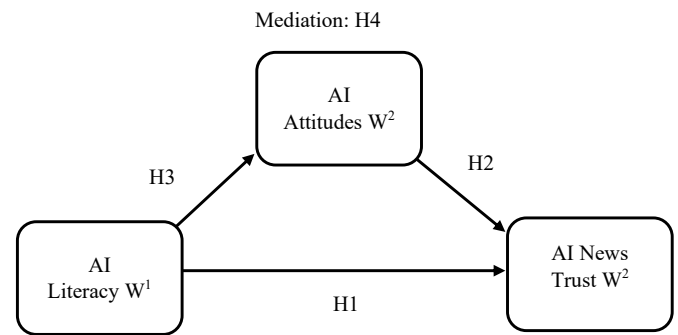


Fig. 1. Conceptual model with hypotheses.

use, attitudes toward artificial intelligence, political orientations, and civic dispositions. For this study, two survey waves were fielded between November 2024 and May 2025: Wave 1 (November 2024) and Wave 2 (February 2025). To ensure data quality and representativeness, [Anonymized] University partnered with YouGov, which provided online panels from opt-in respondents in the United States, Spain, and Chile. YouGov distributed the questionnaire through its proprietary platform and used weighted panel samples designed to approximate national demographic distributions. Sampling quotas were established by key sociodemographic characteristics, including gender, income, and education. Standard attention checks were embedded throughout the instrument to detect careless responding, reducing possible biases common in online panels. The demographic distribution of the Wave 1 sample across countries, together with comparable survey benchmarks, is reported in the Online Appendix (Tables A1–A3).

Participants failing attention checks or completing the questionnaire in under 10 min, both thresholds determined insufficient for meaningful engagement, were excluded from analysis. In total, 115 respondents (1.06%) were removed in Wave 1 (U.S. = 44 [1.20%]; Spain = 50 [1.39%]; Chile = 21 [0.59%]) and 51 respondents (0.93%) in Wave 2 (U.S. = 5 [0.27%]; Spain = 31 [1.68%]; Chile = 15 [0.83%]).

Missing data were minimal across survey waves. The highest item-level missingness in Wave 1 was 0.44% for the U.S., 1.04% for Spain, and 2.31% for Chile; in Wave 2, maximum missingness declined to 0.39%, 0.28%, and 1.44%, respectively. The vast majority of variables had fewer than ten missing observations (U.S. = 98.39%; Spain = 97.87%; Chile = 92.99% in Wave 1; and U.S. = 99.73%; Spain = 99.19%; Chile = 98.37% in Wave 2).

Completion rates for the first wave averaged 79.9% (U.S. = 81.19%; Spain = 84.42%; Chile = 72.75%). Cooperation between Waves 1 and 2 was 50.70% overall, with country-specific rates at U.S. = 49.51%, Spain = 51.21%, and Chile = 51.43%. The final sample comprised respondents from the United States ($N = 1815$), Spain ($N = 1811$), and Chile ($N = 1802$). Unless otherwise specified, survey items used 10-point Likert-type response formats.

Waves were separated by a three-month interval. Scale reliability was evaluated through both internal consistency and test–retest stability. Cronbach's alpha coefficients across the two waves were satisfactory (see measures section). Furthermore, the intraclass correlation coefficient for trust in AI news derived from a two-way mixed effects model assessing absolute agreement yielded ICC = .529, 95% CI [.495, .561], $F(1796, 1796) = 3.25$, $p < .001$, indicating a moderate level of test–retest reliability. Similarly, the ICC for AI attitudes across waves yielded a moderate-to-good level of stability, ICC = .735, 95% CI [.712, .756], $F(1791, 1791) = 6.582$, $p < .001$. The data reported in the following section is based on the matched sample for each country.

2.1. Measurements

Artificial intelligence literacy. Based on prior operationalizations of the

construct (Carolus et al., 2023; Long & Magerko, 2020), this measurement assesses individuals' self-perceived, practical, and operational understanding of artificial intelligence. Specifically, it captures basic comprehension, recognition skills, functional application, and perceived performance benefits of AI in people's daily lives. Participants were asked to indicate their level of agreement (1 = *Strongly disagree* to 10 = *Strongly agree*) with the following items: "I know how AI can help me," "I can skillfully use AI applications or products to help me with my daily work," "I can use AI applications and products to improve my work efficiency," and "I can identify the AI employed in the applications and products I use". Items were averaged to create an artificial intelligence literacy index (three countries: [W¹] $M = 4.56$, $SD = 2.64$, Cronbach's $\alpha = .93$; USA: [W¹] $M = 4.38$, $SD = 2.70$, Cronbach's $\alpha = .93$; Spain: [W¹] $M = 4.37$, $SD = 2.53$, Cronbach's $\alpha = .93$; Chile: [W¹] $M = 4.95$, $SD = 2.54$, Cronbach's $\alpha = .92$).

Artificial intelligence attitudes. This construct captures individuals' general attitudes toward AI, including emotional responses, trust and perceived integrity, and behavioral intentions. Adapting and extending prior measurements of the construct (Sindermann et al., 2021), participants' attitudes toward artificial intelligence (AI) were assessed by asking them to indicate their level of agreement (1 = *Strongly disagree* to 10 = *Strongly agree*) with the following statements: "I generally fear AI" (reverse coded), "I feel generally uneasy about AI" (reverse coded), "I don't know why, but AI scares me" (reverse coded), "AI is dependable," "I trust the performance of AI," "I expect to use AI in the future," "I would recommend AI to my friends and family," "Overall, AI will harm humankind" (reverse coded), "AI is deceptive" (reverse coded), and "AI has integrity." Items were averaged to create an index of artificial intelligence attitudes (three countries: [W¹] $M = 5.32$, $SD = 1.87$, Cronbach's $\alpha = .89$; [W²] $M = 5.26$, $SD = 1.85$, Cronbach's $\alpha = .89$; USA: [W¹] $M = 5.16$, $SD = 1.99$, Cronbach's $\alpha = .91$; [W²] $M = 5.09$, $SD = 1.94$, Cronbach's $\alpha = .91$; Spain: [W¹] $M = 5.28$, $SD = 1.75$, Cronbach's $\alpha = .89$; [W²] $M = 5.28$, $SD = 1.76$, Cronbach's $\alpha = .90$; Chile: [W¹] $M = 5.52$, $SD = 1.85$, Cronbach's $\alpha = .88$; [W²] $M = 5.40$, $SD = 1.82$, Cronbach's $\alpha = .87$). Prior to the analysis, all negatively worded items were reverse-coded so that higher values consistently represented more positive attitudes toward AI.

Artificial intelligence news trust. This construct measures the extent to which individuals perceive political and public affairs news generated by artificial intelligence as trustworthy and accurate. Specifically, it measures participants' level of agreement (1 = *Strongly disagree* to 10 = *Strongly agree*) with the following items: "I trust news on politics and public affairs created by AI," "News created by AI is accurate". Items were averaged to create an index of artificial intelligence news trust (three countries: [W¹] $M = 3.32$, $SD = 2.30$, Spearman-Brown Coefficient = .92; [W²] $M = 3.42$, $SD = 2.33$, Spearman-Brown Coefficient = .91; USA: [W¹] $M = 3.20$, $SD = 2.34$, Spearman-Brown Coefficient = .93; [W²] $M = 3.25$, $SD = 2.33$, Spearman-Brown Coefficient = .92; Spain: [W¹] $M = 3.36$, $SD = 2.27$, Spearman-Brown Coefficient = .92; [W²] $M = 3.50$, $SD = 2.28$, Spearman-Brown Coefficient = .91. Chile: [W¹] $M = 3.40$, $SD = 2.29$, Spearman-Brown Coefficient = .90; [W²] $M = 3.51$, $SD = 2.37$, Spearman-Brown Coefficient = .91).

2.2. Controls

In order to isolate the direct and indirect effects of the variables of interest on the dependent variable, several control variables identified as relevant by prior literature (Araujo et al., 2020) were included in the analysis. Specifically, demographic characteristics (age [Pooled sample, $M = 50.91$, $SD = 15.83$; USA, $M = 53.76$, $SD = 17.74$; Spain, $M = 50.74$, $SD = 13.72$; Chile, $M = 48.22$, $SD = 15.29$], gender [Pooled sample, male = 48.1%; USA, male = 48.8%; Spain, male = 48.9%; Chile, male = 46.5%], education [USA, median = some college; Spain, median = high school graduate; Chile, median = some college], ethnicity [Pooled sample, White-Caucasian = 82%; USA, White-Caucasian: 73.3%; Spain, White-Caucasian = 91.7%; Chile, White-Caucasian = 81.6%], and

income [USA, median = \$30,000 to \$49,999; Spain, median = 20,000 to 29,999 €; Chile, median = 12 to 20,000,000 pesos]), political orientations (political interest ["How interested are you in information about what's going on in politics and public affairs?" and "How closely do you pay attention to information about what's going on in politics and public affairs?", 1 = *Not at all* to 10 = *Very closely*], and political ideology ["Generally speaking, do you usually think of yourself as a Republican or a Democrat?", 0 = *Strong democrat* (or the equivalent for each country) to 10 = *Strong Republican* (or the equivalent for each country), "Where would you place yourself on economic issues?", 1 = *Strong liberal* (or the equivalent for each country) to 11 = *Strong Republican* (or the equivalent for each country) "Where would you place yourself on political issues", 1 = *Strong liberal* (or the equivalent for each country) to 11 = *Strong Republican* (or the equivalent for each country)], media orientations (news use [In the last months, how often did you get news from the following sources: "Local news", "National newspapers", "Online news sites", "Radio news", "Television", 1 = *Never* to 10 = *All the time*]), social media news use [Thinking of the social media you use the most, how often do you use it for the following activities? "To stay informed about current events and public affairs", "To stay informed about my local community", "To get news about current events from mainstream media", 1 = *Never* to 10 = *All the time*], artificial intelligence news use [In the past two months, how frequently have you ... "Used AI to stay informed about current events and public affairs?", "used AI to get additional information about what's going on in politics and public affairs?", and In the past two months, how often did you get news from -AI-based systems (e.g., Gemini, ChatGPT, Copilot, Alexa, Perplexity, etc)", 1 = *Never* to 10 = *All the time*]), and trust (trust in mainstream media [How much do you generally trust news ... "That comes from mainstream news media (e.g., newspapers, TV newscasts, online news sites)?"], 1 = *Not at all* to 10 = *Completely*]) were included as controls in both the autoregressive regression models and indirect effect models. All these control variables, beyond trust in mainstream media and demographic factors, were averaged to create composite constructs for all countries (see Table A4 in the Online Appendix for descriptives and reliabilities).

3. Statistical analyses

Two main statistical approaches were implemented to test the proposed hypotheses: autoregressive regressions and mediation analyses. Both approaches were applied independently for each country, as well as for the pooled dataset across all countries. Specifically, Hypotheses 1 to 3 were tested using a series of autoregressive regressions, while Hypothesis 4 was tested using the PROCESS macro for SPSS (Hayes, 2017).

The series of autoregressive regressions allowed for the establishment of robust time-ordered effects by considering the independent variable at Time 1 and testing its influence on the dependent variable at Time 2, while controlling for baseline levels of the dependent variable and the different potential confounders. Although this technique does not provide causal inference as strongly as experimental designs, it is widely used in public opinion research to provide more robust evidence of temporal effects than cross-sectional analyses. The key advantage of autoregressive models lies in their ability to control for baseline levels of the dependent variable; thus, even a small effect of the independent variable (at Time 1) on the dependent variable (at Time 2) may be meaningful, as most of the variance is already accounted for by the baseline measure (Roth & MacKinnon, 2013).

In addition, to test the proposed hypotheses (and as outlined in the measurements section) a series of control variables were included in the regression models: demographics, political orientation, media orientation, and trust. These variables were entered in separate blocks to assess their individual explanatory power. Furthermore, the final conceptual model depicted in Fig. 1 was broken down into specific hypotheses to construct a coherent theoretical account of how news trust is indirectly influenced. To test the proposed indirect mechanism, Model 4 of the

PROCESS macro for SPSS (Hayes, 2017) was employed.

4. Results

Hypothesis 1 predicted that higher artificial intelligence literacy at W^1 is positively associated with increased artificial intelligence news trust at W^2 in a) the pooled sample, b) the USA, c) Spain, and d) Chile. Results of the autoregressive regressions, reported in Tables 1 and 2, showed a non-statistically significant association in a) the pooled sample, b) the USA, c) Spain, and d) Chile, thus not supporting H1.

In a) the pooled sample, beyond the mediator and baseline levels of the dependent variable ($B = .39, p < .001$), income ($B = .03, p < .05$), political ideology ($B = .04, p < .05$), news use ($B = .04, p < .01$), social media news use ($B = .05, p < .001$), and AI news use ($B = .17, p < .001$) were statistically significant and positive predictors of AI news trust at W^2 . Education ($B = -.03, p < .01$) and political interest ($B = -.014, p < .01$) were statistically significant and negative predictors. Overall, the model predicted 42.2% of the AI news trust W^2 variance in the pooled sample.

In b) the USA, beyond the mediator and baseline levels of the dependent variable ($B = .42, p < .001$), political ideology ($B = .03, p < .05$) and AI news use ($B = .20, p < .001$) were the only statistically significant and positive predictors of AI news trust at W^2 . Overall, the model predicted 53.8% of the AI news trust W^2 variance in the USA.

In the case of c) Spain, beyond the mediator and baseline levels of the dependent variable ($B = .38, p < .001$), political ideology ($B = .06, p < .05$) and AI news use ($B = .14, p < .001$) were the only statistically significant and positive predictors of AI news trust at W^2 . Overall, the model predicted 58.6% of the AI news trust W^2 variance in Spain.

Table 1

Autoregressive model predicting AI attitudes and AI news trust in the pooled countries and in the USA.

	Pooled Countries		USA	
	AI Attitudes W^2	AI News Trust W^2	AI Attitudes W^2	AI News Trust W^2
<i>Block 1: Autoregressive Term</i>				
AI Attitudes	.71***	–	.73***	–
AI News Trust	–	.39***	–	.42***
ΔR^2	60.8%	36.4%	69.6%	47.7%
<i>Block 2: Demographics</i>				
Age	-.03**	-.02	-.03*	-.02
Gender	-.02*	-.02	-.01	.02
Education	.02	-.03**	.00	-.03
Ethnicity	.02*	-.01	-.00	.00
Income	.02*	.03*	-.02	.01
ΔR^2	0.5%	1.1%	0.7%	1.7%
<i>Block 3: Political Orientations</i>				
Political Interest	-.00	-.04**	-.02	-.04
Political Ideology	.02*	.04***	.05***	.03*
ΔR^2	0.1%	0.2%	0.3%	0.1%
<i>Block 4: Media Orientations</i>				
News Use	.00	.04**	-.02	.04
Social Media	.00	.05***	.00	.03
News Use	.04***	.17***	.05*	.20***
ΔR^2	0.3%	3%	0.3%	3.2%
<i>Block 5: Trust Orientations</i>				
Trust in Mainstream Media	-.01	.03*	.02	.02
ΔR^2	0%	0.1%	0%	0.1%
<i>Block 6: Mediator</i>				
AI Attitude	–	.14***	–	.13***
ΔR^2	–	1.5%	–	1.1%
<i>Block 7: Variable of Interest</i>				
AI Literacy	.07***	.01	.10***	.03
ΔR^2	0.3%	0%	0.3%	0%
Total R^2	61.9%	42.2%	71.3%	53.8%
Adjs. R^2	61.9%	42%	71.1%	53.5%
Residual Std. Error	1.14	1.78	1.04	1.59

Note. Sample size for the pooled countries: AI attitudes ($n = 5430$), AI news trust ($n = 5155$); USA: AI attitudes ($n = 1816$), AI news trust ($n = 1798$). Betas are standardized coefficients from the final model. * $p < .05$, ** $p < .01$, *** $p < .001$.

Table 2

Autoregressive model predicting AI attitudes and AI news trust in Spain and Chile.

	Spain		Chile	
	AI Attitudes W^2	AI News Trust W^2	AI Attitudes W^2	AI News Trust W^2
<i>Block 1: Autoregressive Term</i>				
AI Attitudes	.70***	–	.66***	–
AI News Trust	–	.38***	–	.37***
ΔR^2	57.4%	33.2%	53.8%	28.4%
<i>Block 2: Demographics</i>				
Age	.01	-.03	-.03	-.01
Gender	-.01	-.01	-.04*	-.05*
Education	.03*	-.05*	.04*	-.02
Ethnicity	.01	-.04*	.05**	-.01
Income	.03	.02	.01	.05*
ΔR^2	0.4%	0.7%	0.7%	0.9%
<i>Block 3: Political Orientations</i>				
Political Interest	.02	-.02	.00	-.06*
Political Ideology	.03	.06**	-.01	.04
ΔR^2	0.3%	0.9%	0%	0.2%
<i>Block 4: Media Orientations</i>				
News Use	-.03	.03	.01	.04
Social Media	.03	.09***	-.01	.04
News Use	.04	.14***	.03	.16***
ΔR^2	0.4%	2.9%	0.3%	2.4%
<i>Block 5: Trust Orientations</i>				
Trust in Media	.01	.04*	-.03	.02
ΔR^2	0%	0.2%	0.1%	0%
<i>Block 6: Mediator</i>				
AI Attitudes	–	.13***	–	.15***
ΔR^2	–	1.4%	–	1.7%
<i>Block 7: Variable of Interest</i>				
AI Literacy	.06*	.01	.09***	-.01
ΔR^2	0.2%	0%	0.4%	0%
Total R^2	58.6%	39.2%	55.3%	33.7%
Adjs. R^2	58.3%	38.7%	54.9%	33.1%
Residual Std. Error	1.14	1.78	1.22	1.95

Note. Sample size for Spain: AI attitudes ($n = 1811$), AI news trust ($n = 1700$); Chile: AI attitudes ($n = 1803$), AI news trust ($n = 1657$). Betas are standardized coefficients from the final model. * $p < .05$, ** $p < .01$, *** $p < .001$.

.01), social media news use ($B = .09, p < .001$), AI news use ($B = .14, p < .001$), and trust in mainstream media ($B = .04, p < .05$) were statistically significant and positive predictors of AI news trust at W^2 . Education ($B = -.05, p < .05$) and ethnicity ($B = -.04, p < .05$) were statistically significant and negatively associated. Overall, the model predicted 39.2% of the variance in AI news trust at W^2 in Spain.

Finally, in the case of d) Chile, beyond the mediator and baseline levels of the dependent variable ($B = .37, p < .001$), income ($B = .05, p < .05$) and AI news use ($B = .16, p < .001$) were statistically significant and positive predictors of AI news trust at W^2 , while gender ($B = -.05, p < .05$) and political interest ($B = -.06, p < .05$) were statistically significant and negatively associated. Overall, this model accounted for 33.7% of the variance in AI news trust at W^2 .

Hypothesis 2 predicted that more favorable attitudes toward artificial intelligence at W^1 would be positively associated with increased trust in artificial intelligence news at W^2 in a) the pooled sample, b) the USA, c) Spain, and d) Chile. Results of the autoregressive regressions revealed a statistically significant and positive association in a) the pooled sample ($B = .14, p < .001$), b) the USA ($B = .13, p < .001$), c) Spain ($B = .13, p < .001$), and d) Chile ($B = .15, p < .001$), thus supporting H2.

Hypothesis 3 predicted that higher artificial intelligence literacy at W^1 would be positively associated with more favorable attitudes toward artificial intelligence at W^2 in a) the pooled sample, b) the USA, c) Spain, and d) Chile. Results of the autoregressive regressions revealed a statistically significant and positive association in a) the pooled sample ($B = .07, p < .001$), b) the USA ($B = .10, p < .001$), c) Spain ($B = .06, p < .001$), and d) Chile ($B = .10, p < .001$), thus supporting H3.

.05), and d) Chile ($B = .09, p < .001$), thus supporting H3.

In the case of a) the pooled sample, beyond the variable of interest, baseline levels of AI attitudes ($B = .71, p < .001$), ethnicity ($B = .02, p < .05$), income ($B = .02, p < .05$), political ideology ($B = .02, p < .001$), and AI news use ($B = .04, p < .001$) were statistically significant and positive predictors of AI attitudes at W^2 . Age ($B = -.03, p < .01$) and gender ($B = -.02, p < .05$) were statistically significant and negative predictors. The overall model explained 61.9% of the variance in AI attitudes at W^2 .

In the case of b) the USA, beyond the variable of interest, baseline levels of the dependent variable ($B = .73, p < .001$), political ideology ($B = .05, p < .001$), and AI news use ($B = .05, p < .05$) were statistically significant and positive predictors of AI attitudes at W^2 , while only age ($B = -.03, p < .05$) was a statistically significant negative predictor. Overall, the model predicted 71.3% of the variance in the USA.

In c) Spain, beyond the variable of interest, only baseline levels of AI attitudes ($B = .70, p < .001$) and education ($B = .03, p < .05$) were statistically significant and positive predictors of AI attitudes at W^2 . Overall, the model predicted 58.6% of the variance in Spain.

Finally, in d) Chile, beyond the independent variable, baseline levels of the dependent variable ($B = .66, p < .001$) and ethnicity ($B = .05, p < .05$) were statistically significant and positive predictors of AI attitudes at W^2 . Only gender ($B = -.04, p < .05$) was a statistically significant and negative predictor. Overall, the model predicted 55.3% of the variance in Chile.

Finally, Hypothesis 4 predicted that higher artificial intelligence literacy at W^1 would be positively associated with more favorable attitudes toward artificial intelligence at W^2 , which, in turn, would be positively associated with increased trust in artificial intelligence news at W^2 in a) the pooled sample, b) the USA, c) Spain, and d) Chile. Results of the mediation analyses, reported in Figs. 2–5, supported these hypotheses. Specifically, after controlling for the same confounder variables as those reported in Tables 1 and 2 (and also including baseline levels of the dependent variable in the pathway between AI literacy and AI attitudes), the results revealed that in a) the pooled sample (Mediation Effect = .02, 95% CI [LLCI = .01, ULCI = .03]), b) the USA (Mediation Effect = .02, 95% CI [LLCI = .01, ULCI = .04]), c) Spain (Mediation Effect = .01, 95% CI [LLCI = .0002, ULCI = .0288]), and d) Chile (Mediation Effect = .03, 95% CI [LLCI = .01, ULCI = .04]), higher artificial intelligence literacy was associated with more favorable artificial intelligence attitudes, which, in turn, influenced artificial intelligence news trust over time.

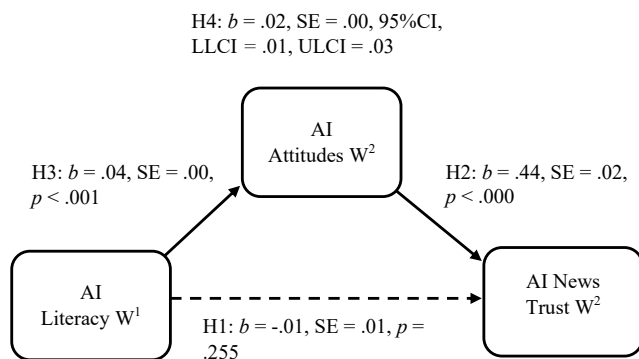


Fig. 2. Indirect and Direct Effect of Artificial Intelligence Literacy W^1 on Artificial Intelligence News Trust W^2 through Artificial Intelligence Attitudes W^2 in the Pooled Sample.

Note. Unstandardized regression coefficients are reported. The model included all control variables used in the autoregressive regressions. However, unlike the autoregressive regressions, this model also controlled for baseline levels of the dependent variable (i.e., artificial intelligence news trust at W^1) when predicting artificial intelligence attitudes at W^2 . The confidence level for the mediation analysis was set at 95%. The number of bootstrap samples used to generate percentile bootstrap confidence intervals was set at 5000.

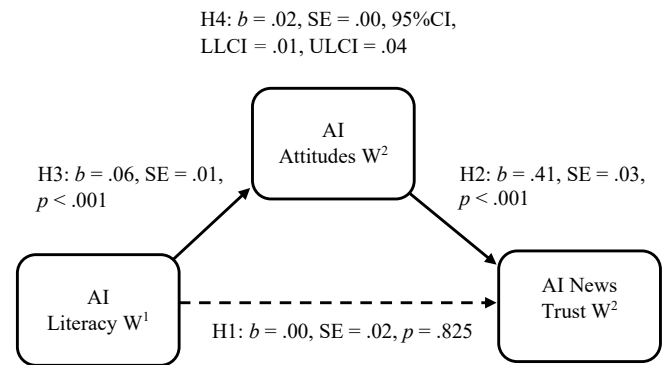


Fig. 3. Indirect and Direct Effect of Artificial Intelligence Literacy W^1 on Artificial Intelligence News Trust W^2 through Artificial Intelligence Attitudes W^2 in the USA.

Note. Unstandardized regression coefficients are reported. The model included all control variables used in the autoregressive regressions. However, unlike the autoregressive regressions, this model also controlled for baseline levels of the dependent variable (i.e., artificial intelligence news trust at W^1) when predicting artificial intelligence attitudes at W^2 . The confidence level for the mediation analysis was set at 95%. The number of bootstrap samples used to generate percentile bootstrap confidence intervals was set at 5000.

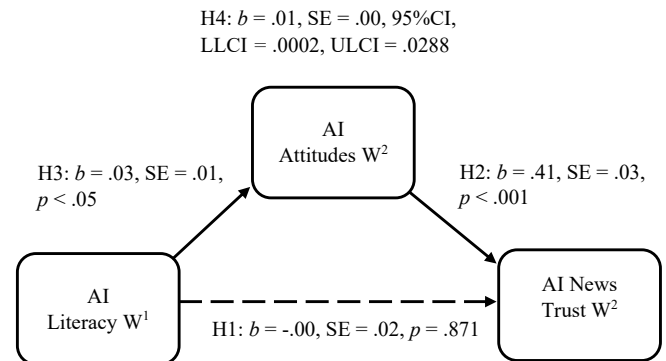


Fig. 4. Indirect and Direct Effect of Artificial Intelligence Literacy W^1 on Artificial Intelligence News Trust W^2 through Artificial Intelligence Attitudes W^2 in the Spain.

Note. Unstandardized regression coefficients are reported. The model included all control variables used in the autoregressive regressions. However, unlike the autoregressive regressions, this model also controlled for baseline levels of the dependent variable (i.e., artificial intelligence news trust at W^1) when predicting artificial intelligence attitudes at W^2 . The confidence level for the mediation analysis was set at 95%. The number of bootstrap samples used to generate percentile bootstrap confidence intervals was set at 5000.

5. Discussion

The aim of this study was to explore the relationships between AI literacy, attitudes towards AI and trust in AI-generated news. We used two waves from a longitudinal study comparing the US, Chile, and Spain to answer this question and explore the generalizability of the findings across countries. We did not find the predicted positive relationship between AI literacy and subsequent trust in AI-generated news, but we found a positive relationship of attitudes and subsequent trust, and between AI Literacy and AI attitudes. More important, there was an indirect effect of AI literacy on trust in AI-generated news in the second wave via AI attitudes. This pattern was stable across all three countries.

One reason why we did not find an effect of AI literacy on trust in AI-generated news one wave later could be methodological. We controlled for trust in AI-generated news at wave 1 and included many control variables, and this makes it more difficult to find significant effects in autoregressive panel designs (Finkel, 1995). Another theoretical

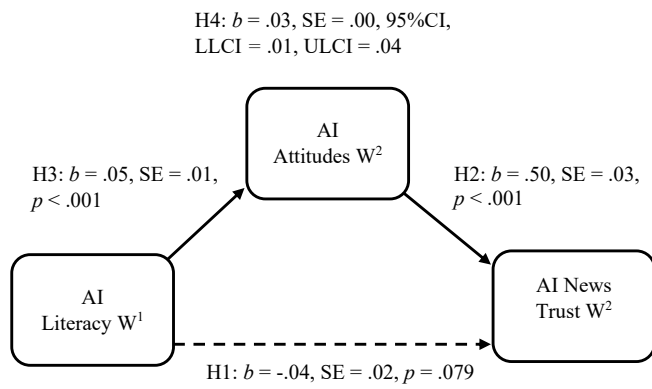


Fig. 5. Indirect and Direct Effect of Artificial Intelligence Literacy W^1 on Artificial Intelligence News Trust W^2 through Artificial Intelligence Attitudes W^2 in Chile.

Note. Unstandardized regression coefficients are reported. The model included all control variables used in the autoregressive regressions. However, unlike the autoregressive regressions, this model also controlled for baseline levels of the dependent variable (i.e., artificial intelligence news trust at W^1) when predicting artificial intelligence attitudes at W^2 . The confidence level for the mediation analysis was set at 95%. The number of bootstrap samples used to generate percentile bootstrap confidence intervals was set at 5000.

explanation is that the association between AI literacy and trust is not straightforward and depends on the dimension of AI literacy use. [Rheu and Cho \(2025\)](#) found differential relationships between understanding the principles of AI and self-efficacy and verification behaviors. In the context of using AI for news, higher literacy might be related to higher bias awareness ([Tully et al., 2025](#)) which should come with lower trust. However, higher literacy could come with better prompting and thus better results (e.g., more personalized, in easier language) which might increase trust. By focusing more on skills and efficacy, we had aimed to shed more light on this second path. Different facts of AI literacy are, however, often positively correlated ([Carolus et al., 2023](#)), so our measure might also be a proxy for the knowledge facets and these two paths might have canceled each other out, resulting in a non-significant relationship when looking at the direct effect. Adding attitude as a mediator and looking at the indirect effect helped us to get a better understanding of the relationship between the skills/efficacy dimension of AI literacy and trust in AI news and this is also the major contribution of our paper. In line with the assumption that higher AI skills relate to a more positive attitude, likely because people make more positive experiences, we found a positive relationship between AI literacy at wave 1 on AI attitudes at wave 2 (even when controlling for attitudes in the first wave and other control variables). We thus contribute to the recent discussion whether AI literacy might not always have beneficial effects ([Rheu & Cho, 2025](#)). Especially the ease of use of LLMs might increase efficiency perceptions and come with the danger of overtrust in AI in domains like news or information search.

Although our auto-regressive models show that attitudes at wave 1 are related to higher trust levels at time 2 when controlling for trust at wave 1 and thus hint that attitudes might cause higher trust level, the additional cross-lagged models showed that this is a reciprocal relationship; higher trust also comes with more positive attitudes in the subsequent wave. Trust has been considered as predictor of attitude frequently as reported in a recent meta-analysis ([Marikyan et al., 2023](#)) and work in related domains has considered attitudes as a mediator between trust and intention to use a technology ([Choung et al., 2023](#); [Ho & Cheung, 2024](#)). Much of this work, is, however, cross-sectional. That we also found a positive relationship between trust and subsequent attitudes, thus, strengthens the argument of these cross-sectional papers that trust might also influence attitudes.

More important is, however, that we should that trust can be an

antecedent but also an effect of attitudes. Our study also sheds light on the dynamic processes between attitudes towards AI and trust in AI-generated news. Moreover, we show that attitudes also mediate the relationship between the skill-dimension of AI literacy and trust in AI-generated news and play, thus, also an important role in building trust.

In the context of classical and alternative news media, [Schwarzenegger \(2020\)](#) found naïve pragmatic trust, when people relied more on their feelings than on knowledge. In the context of trust in AI-news, the sentiment-based path to trust seems to matter.

Interestingly, the basic pattern was the same across all three countries. Thus, although they differ in the levels of AI literacy, attitudes, and trust, with people from Chile descriptively reporting higher literacy, trust, and more positive attitudes than people from Spain and the US, and the countries differ in culture and language (which might affect the performance of AI models), the processes predicting trust are the same, indicating that this is a rather general process. Our results suggest, therefore, that policies focused only on AI literacy may not be enough to increase trust in AI-generated news, because literacy seems to affect trust mainly through people's broader attitudes toward AI. This means that policy efforts should combine skills training with steps that shape how trust is built, such as clear disclosure when AI is used in news production or distribution, and accountability standards for accuracy, sourcing, and corrections.

Despite its contributions, this study is not without limitations. First, the reliance on self-reported measures of AI literacy, attitudes, and trust introduces potential measurement errors. Self-assessments of literacy, in particular, may reflect individuals' perceived rather than actual competence, leading to possible over- or underestimations of their ability to critically evaluate the impact of AI. Second, the literacy scale mainly focused on skillful use but not on awareness of bias or hallucinations. More detailed measures on how exactly people use AI for news (most recent news, regional news, explanations in simpler language) or prompting strategies would have been useful to get more insight into the underlying processes. Third, although the use of a two-wave longitudinal panel allows for more robust inferences about temporal order, it still falls short of establishing strict causality, as experimental designs may accomplish. However, manipulating constructs such as AI literacy and attitudes in experimental settings is quite challenging, as they tend to be relatively stable dispositions that develop gradually through education and experience rather than through short-term exposure or interventions. Taken together, the longitudinal design and the cross-cultural comparison, which demonstrate how the process of AI news trust unfolds, constitute major strengths of our study. These results suggest that our findings are generalizable across countries and contexts, particularly through the inclusion of non-WEIRD nations ([Henrich et al., 2010](#)). To our knowledge, this is the first paper demonstrating an indirect link between AI literacy and trust in AI news via attitudes. This link occurs over time when controlling for autoregressive paths, and in three different countries, indicating that this is a stable pattern that deserves more attention.

CRedit authorship contribution statement

Manuel Goyanes: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Sonja Utz:** Writing – review & editing, Writing – original draft, Conceptualization. **Homero Gil de Zúñiga:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Data curation.

Declaration of informed consent

Informed consent was obtained from all individual participants included in the study. Participants were informed about the purpose of the study, the procedures involved, potential risks and benefits, and their right to withdraw at any time without penalty.

Ethical approval statement

This study was approved by the Institutional Review Board/Ethics Committee of University of Salamanca (Approval IRB #1282). All procedures performed in this study involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments.

Declaration of generative AI use

-AI use: During the preparation of this work, the authors used ChatGPT to improve the clarity and readability of the English language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Funding statement

This study was conducted without any financial support from public, commercial, or not-for-profit funding agencies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chbah.2026.100279>.

References

- Araujo, T., Brosius, A., Goldberg, A. C., Möller, J., & Vreese, C. de. (2023). Humans vs. AI: The role of trust, political attitudes, and individual characteristics on perceptions about automated decision making across Europe. *International Journal of Communication*, 17, 6222–6249.
- Araujo, T., Helberger, N., Kruikemeier, S., & De Vreese, C. H. (2020). In AI we trust? Perceptions about automated decision-making by artificial intelligence. *AI & Society*, 35(3), 611–623.
- Carolus, A., Koch, M. J., Straka, S., Latoschik, M. E., & Wienrich, C. (2023). MAIIS-Meta AI literacy scale: Development and testing of an AI literacy questionnaire based on well-founded competency models and psychological change-and meta-competencies. *Computers in Human Behavior: Artificial Humans*, 1(2), Article 100014.
- Choung, H., David, P., & Ross, A. (2023). Trust in AI and its role in the acceptance of AI technologies. *International Journal of Human-Computer Interaction*, 39(9), 1727–1739. <https://doi.org/10.1080/10447318.2022.2050543>
- Daly, S. J., Wiewiora, A., & Hearn, G. (2025). Shifting attitudes and trust in AI: Influences on organizational AI adoption. *Technological Forecasting and Social Change*, 215, Article 124108. <https://doi.org/10.1016/j.techfore.2025.124108>
- Davis, F. D. (1989). *Technology acceptance model: TAM*. *Information Seeking Behavior and Technology Adoption*, 205(219), 5.
- De Freitas, J., Agarwal, S., Schmitt, B., & Haslam, N. (2023). Psychological factors underlying attitudes toward AI tools. *Nature Human Behaviour*, 7(11), 1845–1854.
- Finkel, S. E. (1995). *Causal analysis with panel data*. Sage.
- Gondwe, G. (2025). *Perceptions of AI-driven news among contemporary audiences: A study of trust, engagement, and impact*. AI & SOCIETY. <https://doi.org/10.1007/s00146-025-02294-x>.
- Guzman, A. L. (2020). Ontological boundaries between humans and computers and the implications for human-machine communication. *Human-Machine Communication*, 1(1), 3.
- Guzman, A. L., & Lewis, S. C. (2020). Artificial intelligence and communication: A Human–Machine Communication research agenda. *New Media & Society*, 22(1), 70–86. <https://doi.org/10.1177/1461444819858691>.
- Hayes, A. F. (2017). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach*. Guilford publications.
- Henrich, J., Heine, S. J., & Norenzayan, A. (2010). The weirdest people in the world? *Behavioral and Brain Sciences*, 33(2–3), 61–83.
- Ho, S. S., & Cheung, J. C. (2024). Trust in artificial intelligence, trust in engineers, and news media: Factors shaping public perceptions of autonomous drones through UTAUT2. *Technology in Society*, 77, Article 102533. <https://doi.org/10.1016/j.techsoc.2024.102533>
- Hornberger, M., Bewersdorff, A., Schiff, D. S., & Nerdel, C. (2025). A multinational assessment of AI literacy among university students in Germany, the UK, and the US. *Computers in Human Behavior: Artificial Humans*, 4, 100132. <https://doi.org/10.1016/j.chbah.2025.100132>.
- Huang, K.-T., & Ball, C. (2024). The influence of AI literacy on user's trust in AI in practical scenarios: A digital divide pilot study. *Proceedings of the Association for Information Science and Technology*, 61(1), 937–939. <https://doi.org/10.1002/pr.a2.1146>.
- Jażwińska, K., & Chandrasekar, A.. *AI search has a citation problem*. Columbia Journalism Review. https://www.cjr.org/tow_center/we-compared-eight-ai-search-engines-they-re-all-bad-at-citing-news.php.
- Kohring, M. (2019). Public Trust in News Media. In T. P. Vos, F. Hanusch, D. Dimitrakopoulou, M. Geertsema-Sligh, & A. Sehl (Eds.), *The International Encyclopedia of Journalism Studies* (pp. 1–6). John Wiley & Sons. <https://doi.org/10.1002/9781118841570.iejs0056>.
- Laupichler, M. C., Aster, A., Meyerheim, M., Raupach, T., & Mergen, M. (2024). Medical students' AI literacy and attitudes towards AI: A cross-sectional two-center study using pre-validated assessment instruments. *BMC Medical Education*, 24(1), 401. <https://doi.org/10.1186/s12909-024-05400-7>.
- Lee, D. K. L., & Ramazan, O. (2021). Fact-checking of health information: The effect of media literacy, metacognition and health information exposure. *Journal of Health Communication*, 26(7), 491–500. <https://doi.org/10.1080/10810730.2021.1955312>
- Lombard, M., & Xu, K. (2021). Social responses to media technologies in the 21st century: The media are social actors paradigm. *Human-Machine Communication*, 2(1), 29–55. <https://doi.org/10.30658/hmc.2.2>.
- Long, D., & Magerko, B. (2020). What is AI literacy? Competencies and design considerations. In *Proceedings of the 2020 CHI conference on human factors in computing systems* (pp. 1–16).
- Lu, D. (2025). January 28). We tried out DeepSeek. It worked well, until we asked it about Tiananmen Square and Taiwan. *The Guardian*. <https://www.theguardian.com/technology/2025/jan/28/we-tried-out-deepseek-it-works-well-until-we-asked-it-about-tiananmen-square-and-taiwan>.
- Marikyan, D., Papagiannidis, S., & Stewart, G. (2023). Technology acceptance research: Meta-analysis. *Journal of Information Science*, Article 01655515231191177.
- Montag, C., & Ali, R. (2025). Can we assess attitudes toward AI with single items? Associations with existing attitudes toward AI measures and trust in ChatGPT. *Journal of Technology in Behavioral Science*. <https://doi.org/10.1007/s41347-025-00481-7>.
- Ng, D. T. K., Leung, J. K. L., Chu, S. K. W., & Qiao, M. S. (2021). Conceptualizing AI literacy: An exploratory review. *Computers and Education: Artificial Intelligence*, 2, Article 100041. <https://doi.org/10.1016/j.caeai.2021.100041>
- Pan, Z., Moore, O. A., Papadimitriou, A., & Zhu, J. (2025). AI literacy and trust: A multi-method study of Human-GAI team collaboration. *Computers in Human Behavior: Artificial Humans*, 4, 100162. <https://doi.org/10.1016/j.chbah.2025.100162>.
- Pinski, M., & Benlian, A. (2024). AI literacy for users – A comprehensive review and future research directions of learning methods, components, and effects. *Computers in Human Behavior: Artificial Humans*, 2(1), Article 100062. <https://doi.org/10.1016/j.chbah.2024.100062>
- Rheu, M.(M. J.), & Cho, J. (2025). The trap of AI literacy: The paradoxical relationships between college students' use of LLMs, AI literacy, and fact-checking behavior. In *Proceedings of the extended abstracts of the CHI conference on human factors in computing systems* (Vol. 25, pp. 1–7). CHI EA. <https://doi.org/10.1145/3706599.3719681>
- Roth, D. L., & MacKinnon, D. P. (2013). Mediation analysis with longitudinal data. In *Longitudinal data analysis* (pp. 181–216). Routledge.
- Ryan, S. (2021). Fake news, epistemic coverage and trust. *The Political Quarterly*, 92(4), 606–612. <https://doi.org/10.1111/1467-923X.13003>.
- Schranz, M., Schneider, J., & Eisenegger, M. (2018). Media trust and media use. In K. Otto, & A. Köhler (Hrsg.), *Trust in Media and Journalism: Empirical Perspectives on Ethics, Norms, Impacts and Populism in Europe* (S. 73–91). Springer Fachmedien. https://doi.org/10.1007/978-3-658-20765-6_5.
- Schwarzenegger, C. (2020). Personal epistemologies of the media: Selective criticality, pragmatic trust, and competence–confidence in navigating media repertoires in the digital age. *New Media & Society*, 22(2), 361–377. <https://doi.org/10.1177/1461444819856919>
- Simon, F., Nielsen, R. K., & Fletcher, R. (2025). Generative AI and news report 2025: How people think about AI's role in journalism and society. <https://reutersinstitute.politics.ox.ac.uk/generative-ai-and-news-report-2025-how-people-think-about-ai-role-journalism-and-society>.
- Sindermann, C., Sha, P., Zhou, M., Wernicke, J., Schmitt, H. S., Li, M., Sariyska, R., Stavrou, M., Becker, B., & Montag, C. (2021). Assessing the attitude towards artificial intelligence: Introduction of a short measure in German, Chinese, and English language. *KI-Künstliche Intelligenz*, 35(1), 109–118.
- Strömbäck, J., Tsifti, Y., Boomgaarden, H., Damstra, A., Lindgren, E., Vliegthart, R., & Lindholm, T. (2020). News media trust and its impact on media use: Toward a framework for future research. *Annals of the International Communication Association*, 44(2), 139–156. <https://doi.org/10.1080/23808985.2020.1755338>
- Sultan, M. I., Bhuiyan, A. J. M. S. A., & Amir, A. S. (2024). Reimagining journalism: Exploring the AI revolution-a thorough analysis of potential advantages and challenges. *Kommunikator*, 16(1), 1–13.
- Sundar, S.S., & Kim, J. (2019). Machine heuristic: When we trust computers more than humans with our personal information. In *Proceedings of the 2019 CHI Conference on human factors in computing systems* (pp. 1–9).
- Tauman Kalai, A., Nachum, O., Vempala, S.S., & Zhang, E. (2025). *Why language models hallucinate*. arXiv. <https://doi.org/10.48550/arXiv.2509.04664>.

Tully, S. M., Longoni, C., & Appel, G. (2025). Lower artificial intelligence literacy predicts greater AI receptivity. *Journal of Marketing*. <https://doi.org/10.1177/00222429251314491> (Sage CA: Los Angeles, CA).

Valtonen, T., Tedre, M., Mäkitalo, K., & Vartiainen, H. (2019). Media literacy education in the age of machine learning. *Journal of Media Literacy Education*, 11(2), 20–36. <https://doi.org/10.23860/jmle-2019-11-2-2>

Vogel, T., & Wanke, M. (2016). *Attitudes and attitude change*. Psychology press.